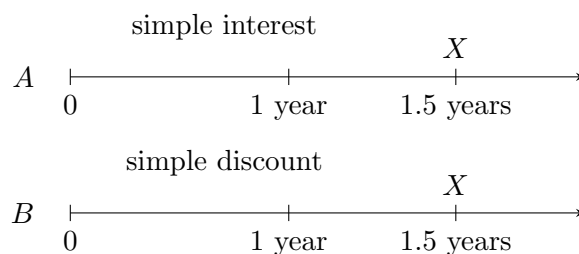


Problem 8.24

Method *A* assumes simple interest over final fractional periods, while Method *B* assumes simple discount over final fractional periods. The annual effective rate of interest is 20%. Find the ratio of the present value of a payment to be made in 1.5 years computed under Method *A* to that computed under Method *B*.

Solution

Let the payment due at time 1.5 be X .



The annual effective interest rate is

$$i = 0.20.$$

Thus the equivalent annual effective discount rate is

$$d = \frac{i}{1+i} = \frac{0.20}{1.20} = \frac{1}{6}.$$

Method A

Method *A* uses simple interest over the final fractional period. Therefore, the present value under Method *A* is

$$\begin{aligned} Y &= X (1 + 0.20(0.5))^{-1} (1.20)^{-1} \\ &= X \left(\frac{1}{1.10} \right) \left(\frac{1}{1.20} \right) \\ &= X \left(\frac{25}{33} \right). \end{aligned}$$

Method B

Method B uses simple discount over the final fractional period. Therefore, the present value under Method B is

$$\begin{aligned} Z &= X (1 - d(0.5)) (1 - d) \\ &= X \left(1 - \frac{1}{6}(0.5)\right) \left(1 - \frac{1}{6}\right) \\ &= X \left(\frac{11}{12}\right) \left(\frac{5}{6}\right) \\ &= X \left(\frac{55}{72}\right). \end{aligned}$$

Hence, the required ratio is

$$\begin{aligned} \frac{Y}{Z} &= \frac{X \left(\frac{25}{33}\right)}{X \left(\frac{55}{72}\right)} \\ &= \frac{120}{121} \\ &\approx 0.9917. \end{aligned}$$